KNEE CONTROLLER

VARIANT: PRELIMINARY

2026-10-16

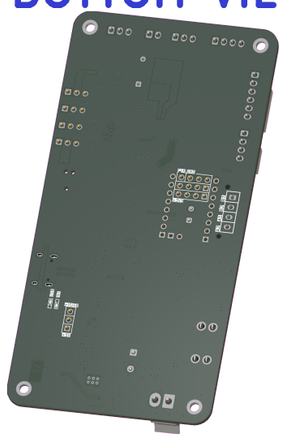
REV A

Page	Index	Page	Index	Page	Index	Page	Index
1	COVER PAGE	11	L1–L2 DRILL MAP	21	.....	31	.....
2	SYSTEM OVERVIEW	12	TOP TESTPOINTS	22	.....	32	.....
3	POWER MANAGEMENT	13	L1 COPPER	23	.....	33	.....
4	MICROCONTROLLER	14	L2 COPPER	24	.....	34	.....
5	MOTOR CONTROL AND IO	15	.....	25	.....	35	.....
6	ASSEMBLY OVERVIEW	16	.....	26	.....	36	.....
7	TOP ASSEMBLY	17	.....	27	.....	37	.....
8	BOTTOM ASSEMBLY	18	.....	28	.....	38	.....
9	TOP FABRICATION	19	.....	29	.....	39	.....
10	BOTTOM FABRICATION	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

MADE WITH LOVE

NOT FITTED COMPONENTS MARKED AS X

DRAFT – EARLY STAGE SCHEMATIC CAPTURE
PRELIMINARY – LATE STAGE SCHEMATIC CAPTURE
CHECKED – PCB LAYOUT CHECKED, CONTACT ENGINEER IF MISTAKE FOUND
RELEASED – READY FOR PRODUCTION

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

REV	DATE	DESCRIPTION	DRAWN	CHECK

SCALE NTS

SCHEMATIC

BOARDWRIGHT

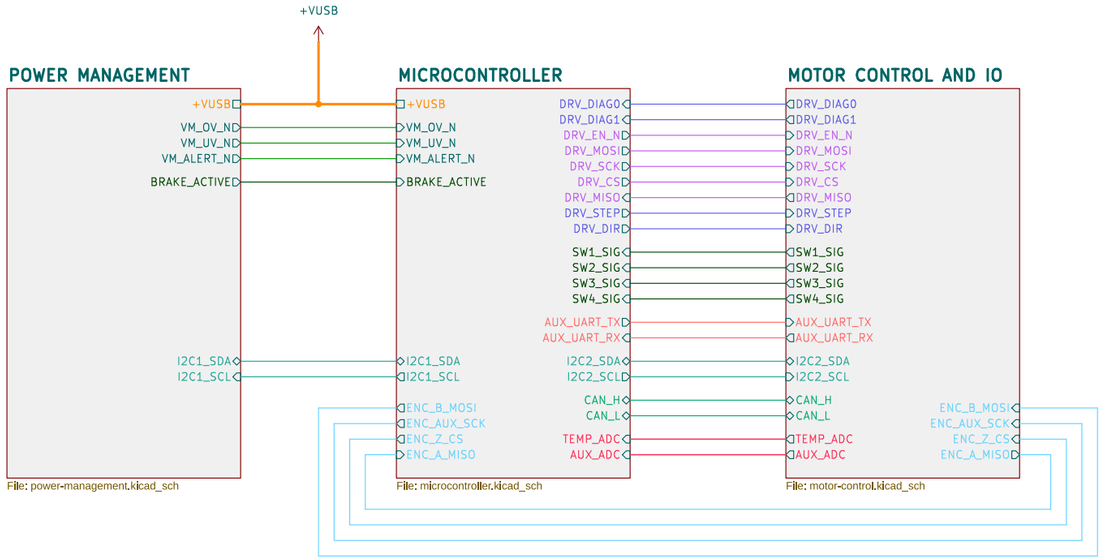
KNEE CONTROLLER

DRG. NO. 4840.703

REV A

A3

SHEET 1 OF 14



REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS

SCHEMATIC

A3

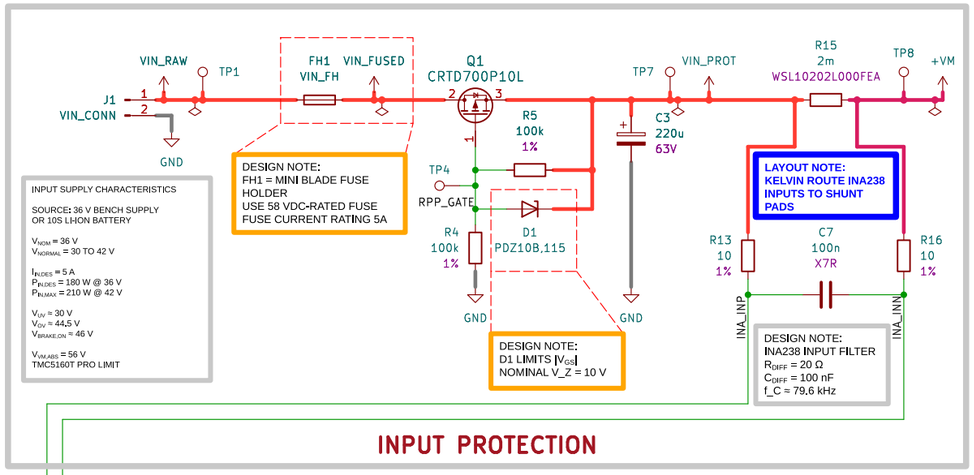
SHEET 2 OF 14

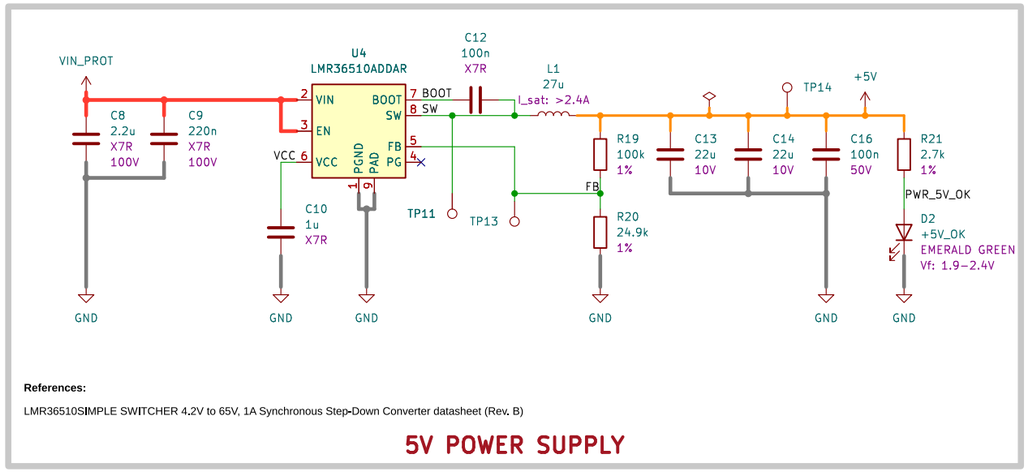
BOARDWRIGHT

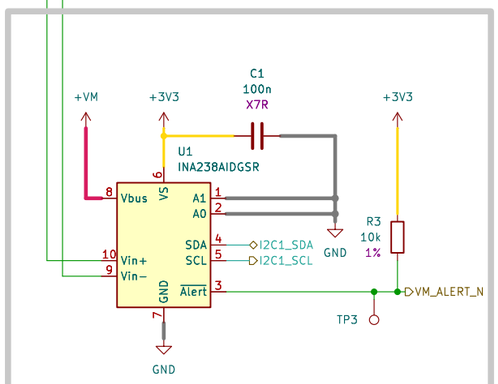
KNEE CONTROLLER

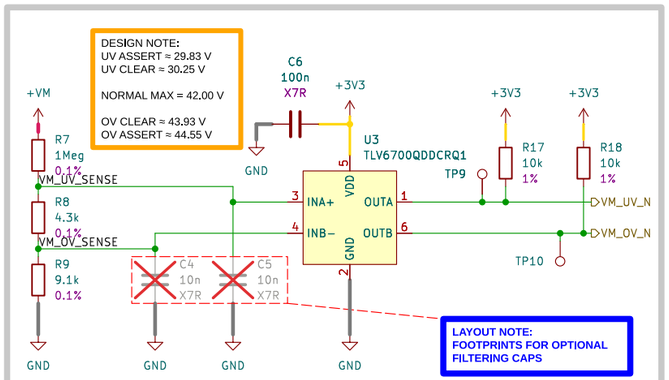
DRG. NO. 4840.703

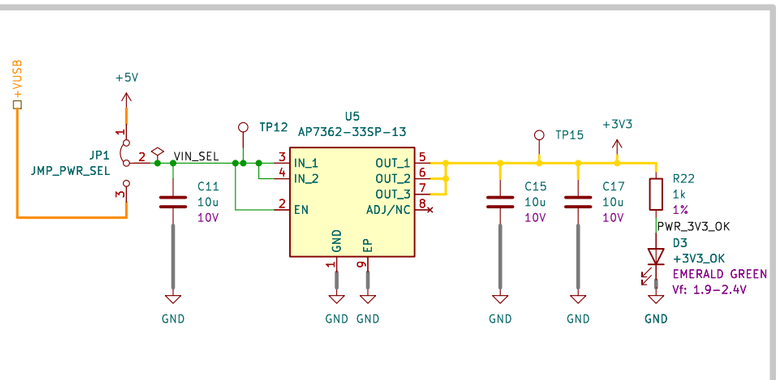
REV A

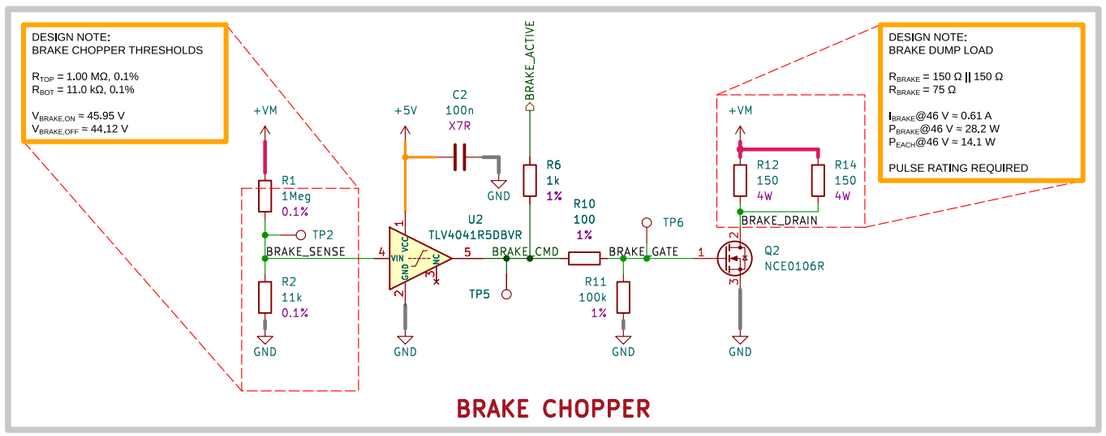












REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS

SCHEMATIC

BOARDWRIGHT

KNEE CONTROLLER

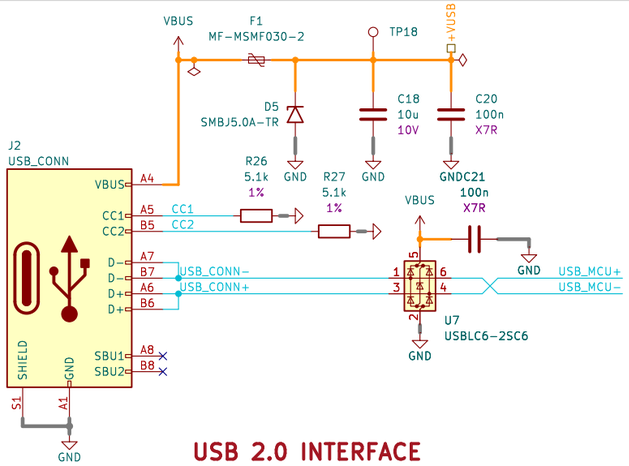
A3

SHEET 3 OF 14

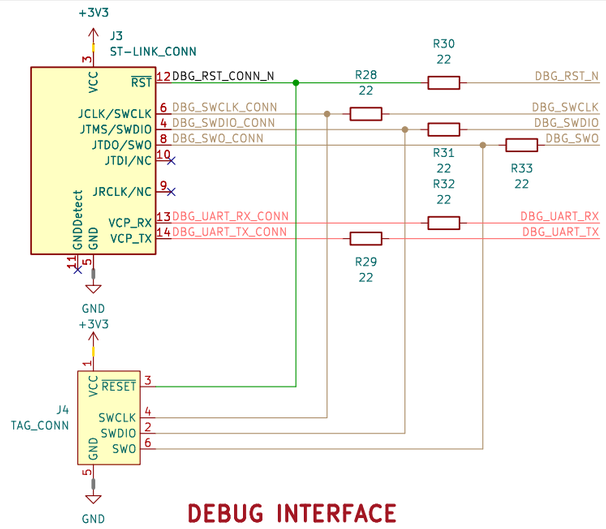
DRG. NO. 4840.703

REV A

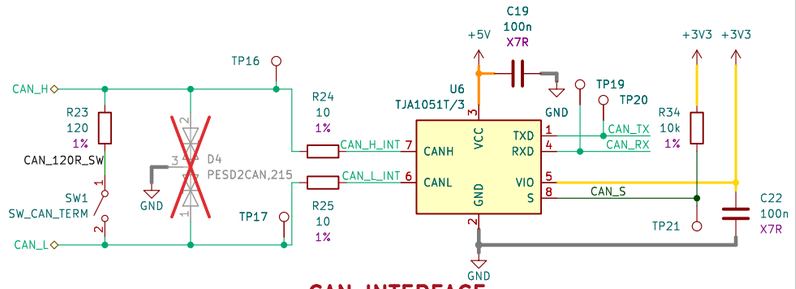
USB 2.0 INTERFACE



DEBUG INTERFACE



CAN INTERFACE



CLOCK SOURCES

DESIGN NOTE:
CRYSTAL LOAD CAPACITORS

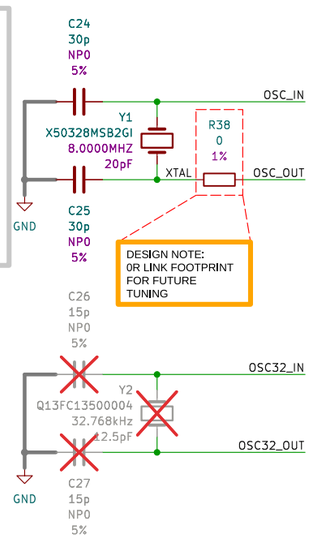
CL = (C1*C2)/(C1+C2) + CS

FOR C1 = C2 = C:
C = 2*(CL - CS)

HSE (Y1):
CL = 20 pF
CS = 10 pF (MCU + PCB EST.)
C24 = C27 = 20 pF
USE C0G/NP0

LSE (Y2):
CL = 12,5 pF
CS = 5 pF (INITIAL EST.)
C33 = C34 = 15 pF
USE C0G/NP0

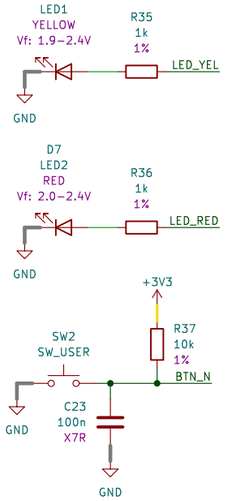
VALUES ARE INITIAL DESIGN VALUES.
VERIFY/TUNE ON FINAL PCB IF
FREQUENCY ACCURACY REQUIRES.



References:

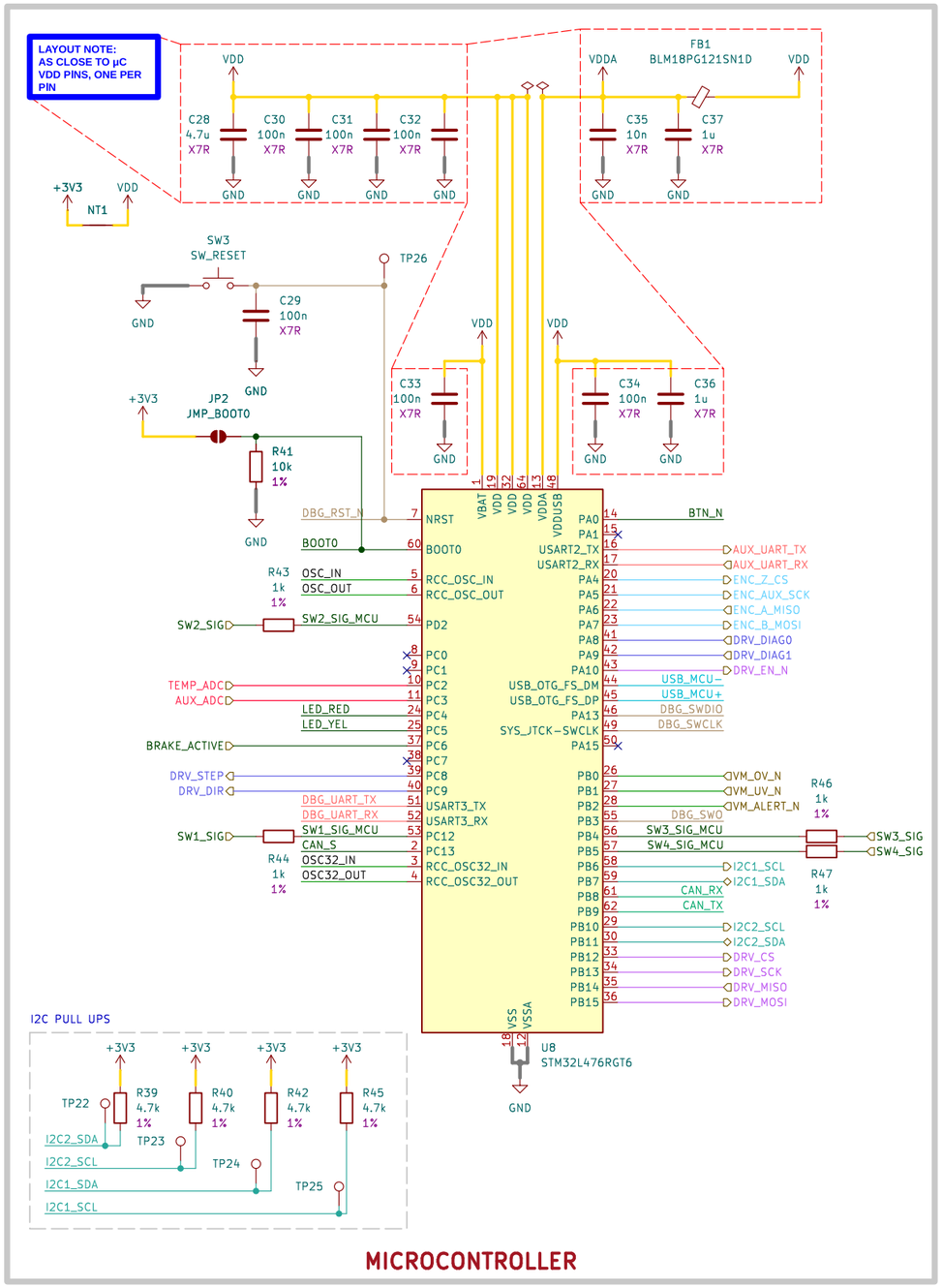
AN2867 - Guidelines for oscillator design on STM8AF/AL/S and STM32 MCUs/MPUs

USER INTERFACE



MICROCONTROLLER

LAYOUT NOTE:
AS CLOSE TO µC
VDD PINS, ONE PER
PIN



I2C PULL UPS

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

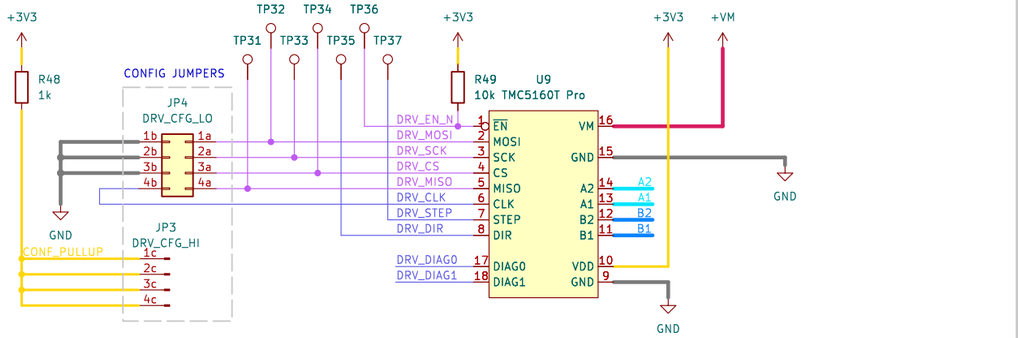
BOARDWRIGHT

SCHEMATIC

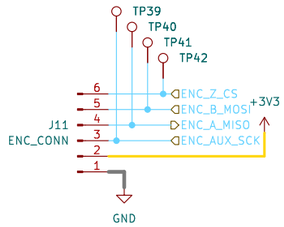
KNEE CONTROLLER

SCALE	DRG. NO.	REV
NTS	4840.703	A
A3	SHEET 4 OF 14	

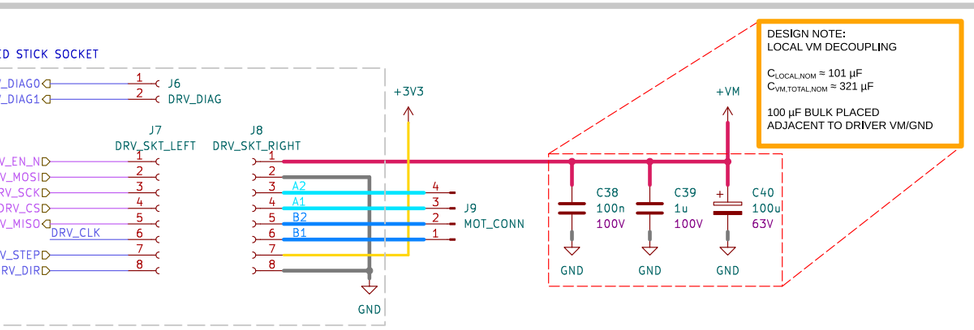
REV	DATE	DESCRIPTION	DRAWN	CHECK



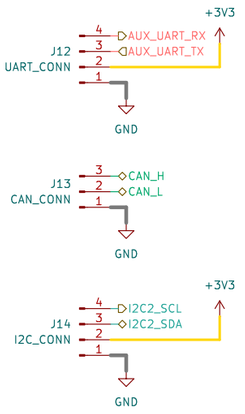
MOTOR DRIVER



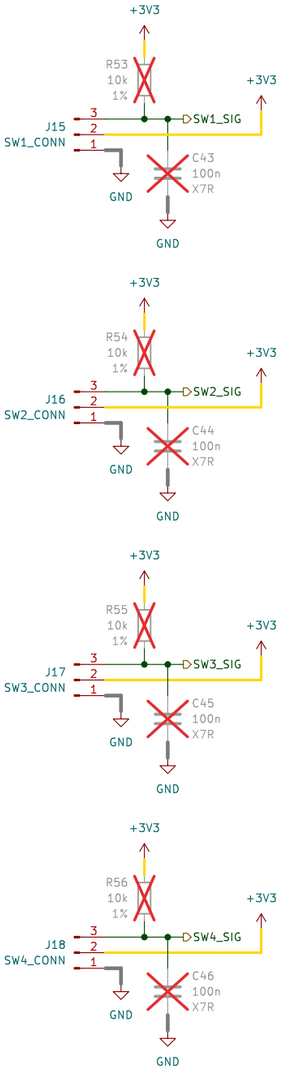
ENCODER INTERFACE



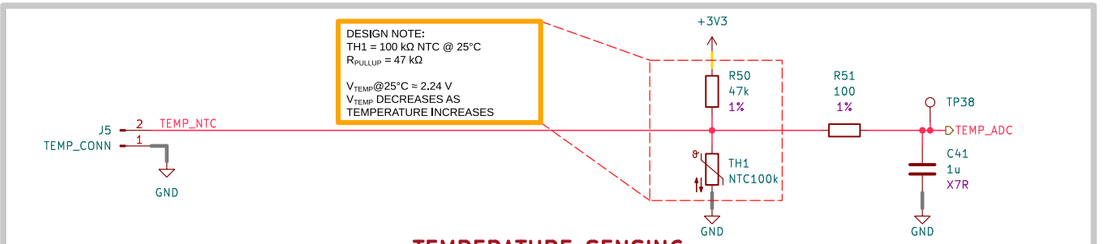
MOTOR INTERFACE



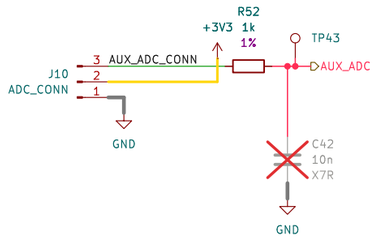
COMMUNICATION INTERFACES



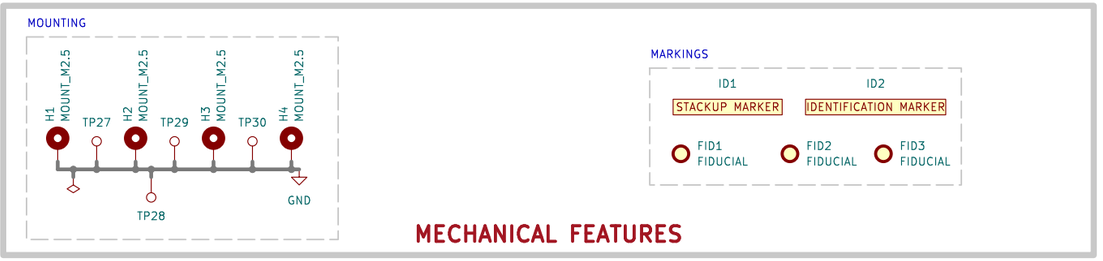
DIGITAL INPUTS



TEMPERATURE SENSING



ANALOG INPUT



MECHANICAL FEATURES

REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS

SCHEMATIC

BOARDWRIGHT

KNEE CONTROLLER

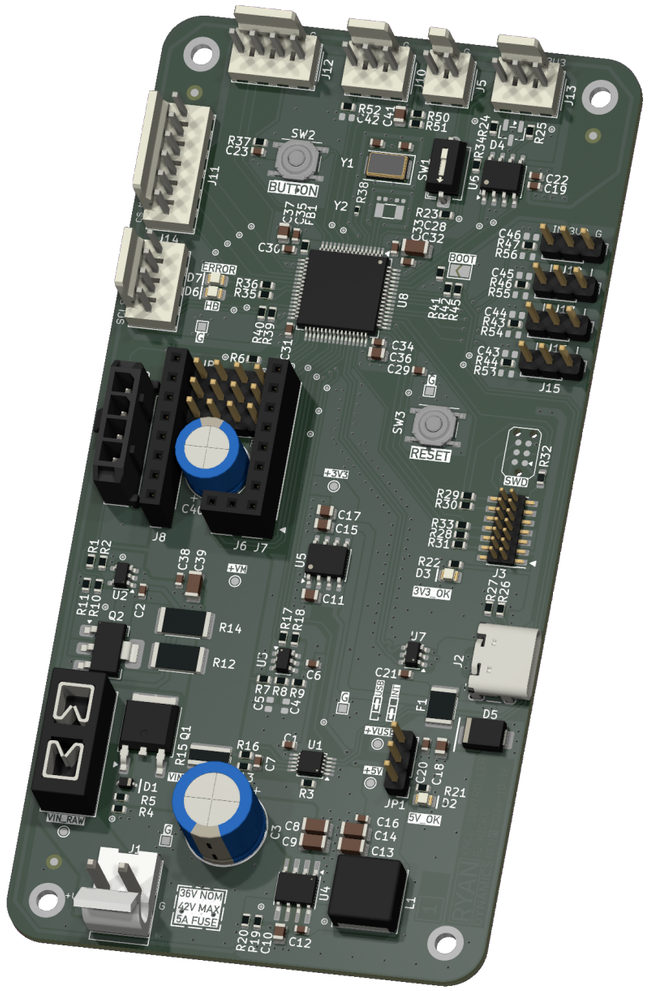
DRG. NO. 4840.703

REV A

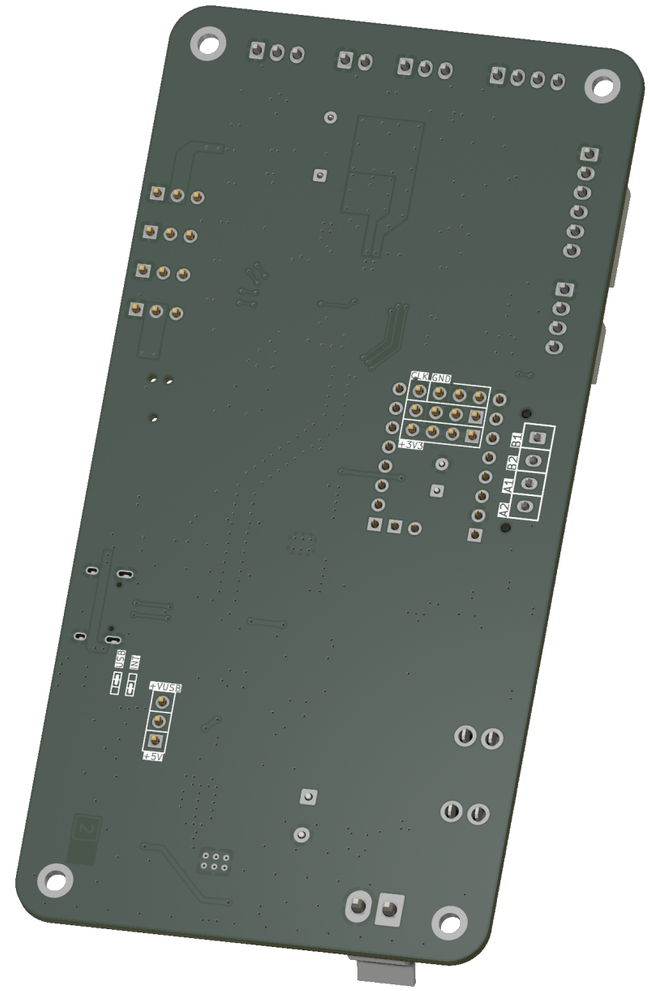
A3

SHEET 5 OF 14

TOP VIEW



BOTTOM VIEW



REV	DATE	DESCRIPTION	DRAWN	CHECK	

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

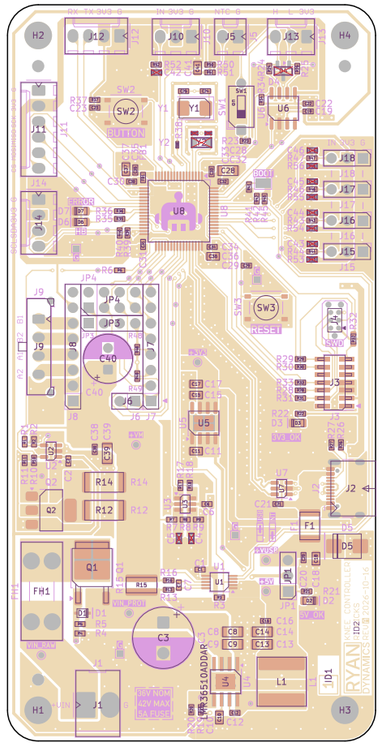
SCALE NTS	ASSEMBLY OVERVIEW	**BOARDWRIGHT**
		KNEE CONTROLLER
A3	SHEET 6 OF 14	DRG. NO. 4840.703 / REV A

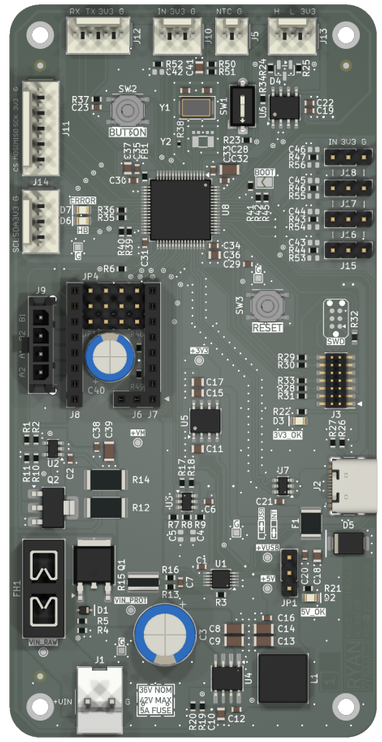
COMPONENTS

Type	Front Side	Back Side	Total
THT	26	0	26
SMT	178	0	178
Total	204	0	204

ASSEMBLY NOTES (UNLESS OTHERWISE SPECIFIED)

1) DO NOT POPULATE PARTS ARE MARKED WITH A RED CROSS.
2) DO NOT POPULATE PARTS ARE NOT PRESENT IN THE BOM.
3) IF CONFLICTING INFORMATION IS FOUND BETWEEN THE ASSEMBLY FILE AND BOM, BOM SHOULD BE USED AS THE MAIN SOURCE.
4) CONFORMAL COATING IS NOT REQUIRED.
5) DOT IDENTIFIES PIN #1 LOCATION AND DEVICE ORIENTATION WHEN VIEWED FROM THE TOP.

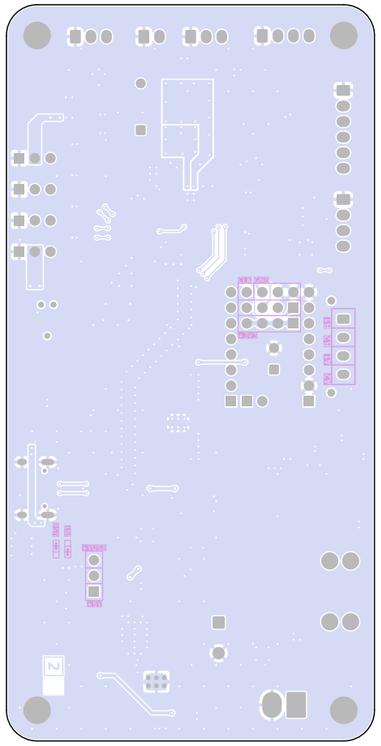


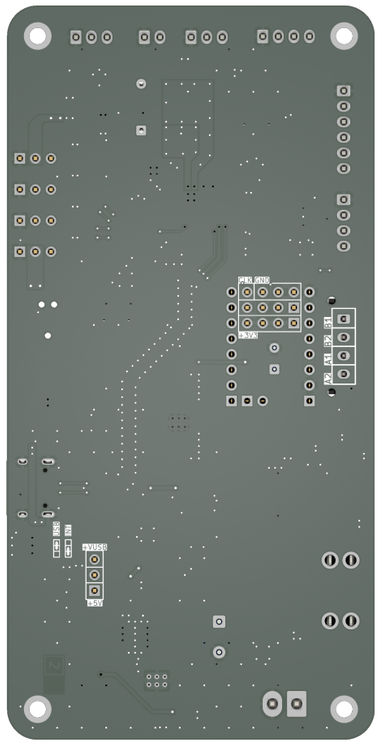


REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS	TOP ASSEMBLY	**BOARDWRIGHT**
		KNEE CONTROLLER
A3	SHEET 7 OF 14	DRG. NO. 4840.703 / REV A





	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

BOARDWRIGHT

REV	DATE	DESCRIPTION	DRAWN	CHECK

SCALE NTS

BOTTOM ASSEMBLY

KNEE CONTROLLER

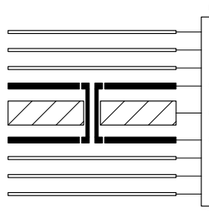
A3

SHEET 8 OF 14

DRG. NO. 4840.703

REV A

LAYER STACKUP LEGEND

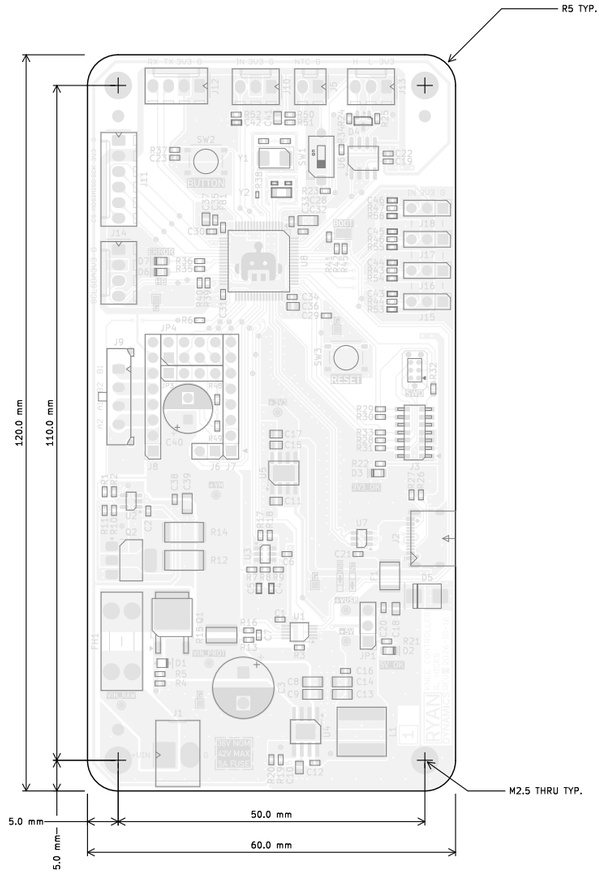


Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0,02mm	Solder Reslst	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0,035mm (1,00oz)**		**Signal**	**GBR**
Core		1,49mm	FR4_7628	Dielectric	
Copper	**L2 (Sig, GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
	B.Mask	0,02mm	Solder Reslst	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1,6mm
Note: external layer thicknesses are specified after plating

IMPEDANCE TABLE

NO IMPEDANCE CONTROLLED TRACES



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) BOARD OUTLINE IS DEFINED IN THE SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF THE CIRCUMSCRIBED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

2) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SURFACE FINISH: HAL LEAD-FREE

4) SOLDER MASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.

5) SILKSCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

6) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDER MASK IS OPENED IN THE GERBER DATA.

7) PCB MATERIAL REQUIREMENTS:

   A. CORE MATERIAL: FR-4.

8) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	60.000 x 120.000 mm
BOARD THICKNESS	1.600 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	0.650 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

9) CONFORMAL COATING IS NOT REQUIRED.

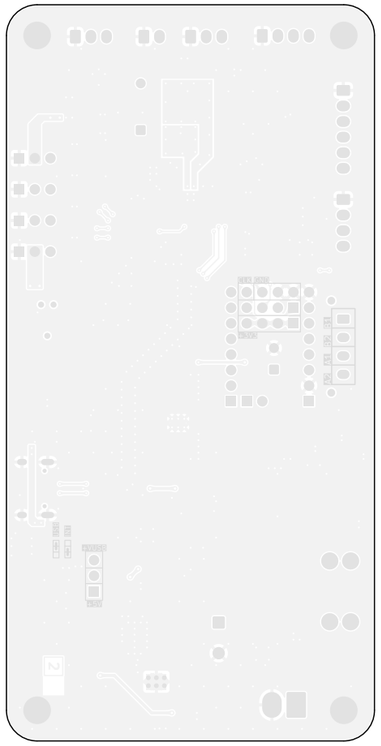
ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

REV	DATE	DESCRIPTION	DRAWN	CHECK

BOARDWRIGHT

SCALE NTS	TOP FABRICATION	KNEE CONTROLLER	
A3	SHEET 9 OF 14	DRG. NO. 4840.704	REV A



ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

BOARDWRIGHT

BOTTOM FABRICATION

KNEE CONTROLLER

SCALE NTS

A3

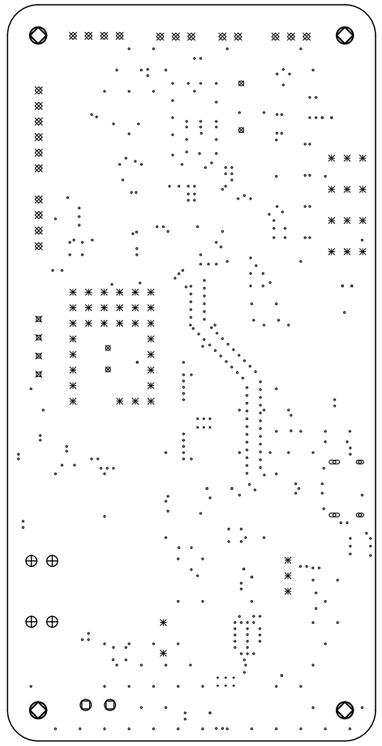
SHEET 10 OF 14

DRG. NO. 4840.704

REV A

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	6	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
○	391	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
+	6	0,33mm (12,99mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
□	11	0,40mm (15,75mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
◇	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	4	0,80mm (31,50mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
✳	47	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	4	1,02mm (40,16mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	22	1,19mm (46,85mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊕	4	1,78mm (70,08mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
○	2	1,80mm (70,87mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
○	4	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
	Total 505					



REV	DATE	DESCRIPTION	DRAWN	CHECK

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

SCALE NTS

L1-L2 DRILL MAP

A3

SHEET 11 OF 14

BOARDWRIGHT

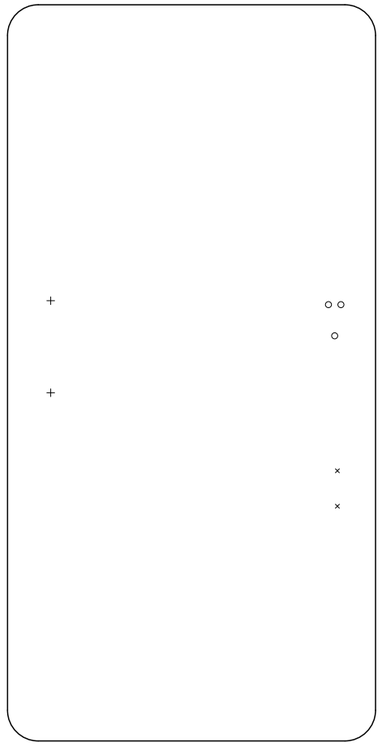
KNEE CONTROLLER

DRG. NO. 4840.704

REV A

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0,65mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
○	3	0,99mm (39,00mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
+	2	1,27mm (50,00mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
	Total 7					



REV	DATE	DESCRIPTION	DRAWN	CHECK

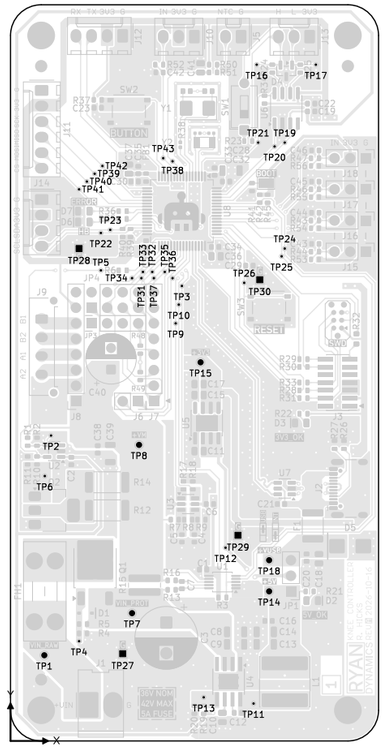
	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

BOARDWRIGHT

SCALE NTS	L1–L2 DRILL MAP
A3	SHEET 11 OF 14

KNEE CONTROLLER

DRG. NO.	REV
4840.704	A



Ref.	Net	X [mm]	Y [mm]
TP1	VIN_RAW	5,51	13,97
TP2	BRAKE_SENSE	6.60	49,78
TP3	VM_ALERT_N	27.94	74,17
TP4	RPP_GATE	11.18	16.26
TP5	BRAKE_CMD	14.73	76.71
TP6	BRAKE_GATE	5,59	43,18
TP7	VIN_PROT	19.81	20,83
TP8	+VM	20.95	48.26
TP9	VM_UV_N	26.92	68.07
TP10	VM_OV_N	27.43	71.12
TP11	SW	39.62	6.10
TP12	VIN_SEL	35.05	31.50
TP13	FB	31.50	7.11
TP14	+5V	42.16	24.38
TP15	+3V3	30.99	61.72
TP16	CAN_H	40,13	110.24
TP17	CAN_L	49.78	110.24
TP18	+VUSB	42,16	29.46
TP19	CAN_RX	44.70	97.54
TP20	CAN_TX	43.05	96.90
TP21	CAN_S	40.39	97,54
TP22	I2C2_SDA	14.73	82.80
TP23	I2C2_SCL	16.26	83.31
TP24	I2C1_SDA	44.70	80.26
TP25	I2C1_SCL	44,20	78,99
TP26	DBG_RST_N	38.10	74,68
TP27	GND	18.29	14.22
TP28	GND	11.18	80.26
TP29	GND	37.08	33.53
TP30	GND	40,64	75,18
TP31	DRV_MISO	21.34	75.44
TP32	DRV_MOSI	23.11	76.45

Ref.	Net	X [mm]	Y [mm]
TP33	DRV_SCK	21.59	76.45
TP34	DRV_CS	19.81	75.44
TP35	DRV_DIR	25.15	76.45
TP36	DRV_EN_N	26.42	75.44
TP37	DRV_STEP	23.37	75.44
TP38	TEMP_ADC	26.42	94.49
TP39	ENC_AUX_SCK	13.72	92.46
TP40	ENC_A_MISO	12.45	91.19
TP41	ENC_B_MOSI	11.18	89.92
TP42	ENC_Z_CS	14.99	93.73
TP43	AUX_ADC	24.89	95.00

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

REV	DATE	DESCRIPTION	DRAWN	CHECK

SCALE NTS

TOP TESTPOINTS

BOARDWRIGHT

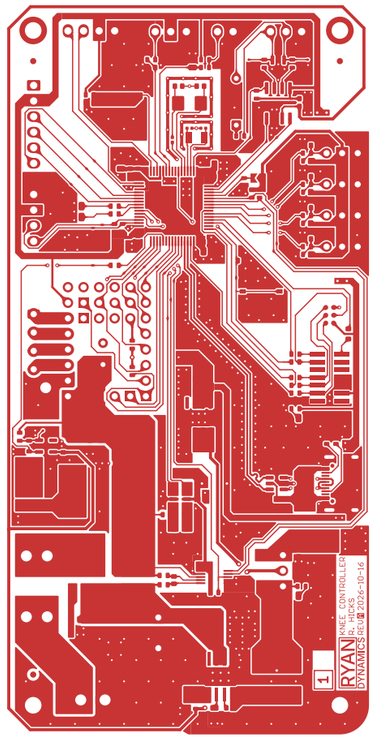
KNEE CONTROLLER

DRG. NO. 4840.704

REV A

A3

SHEET 12 OF 14



	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

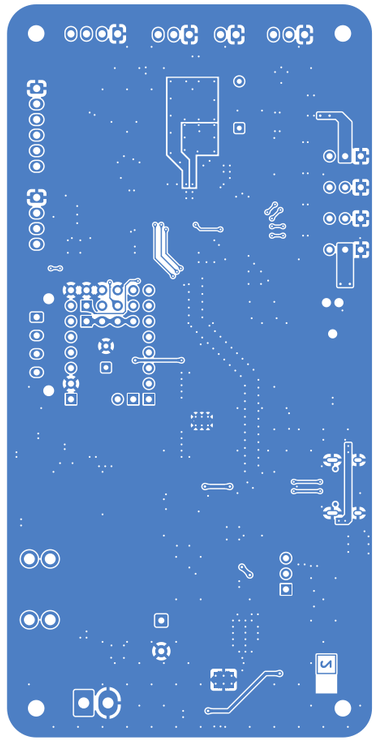
BOARDWRIGHT

L1 COPPER

KNEE CONTROLLER

SCALE	DRG. NO.	REV
NTS	4840.704	A
A3	SHEET 13 OF 14	

REV	DATE	DESCRIPTION	DRAWN	CHECK



	DRAWN	CHECKED	APPROVED
NAME	R. HICKS		
DATE	2026-09-27		

BOARDWRIGHT

REV	DATE	DESCRIPTION	DRAWN	CHECK

SCALE NTS

L2 COPPER

KNEE CONTROLLER

A3

SHEET 14 OF 14

DRG. NO. 4840.704

REV A